A PROJECT REPORT

on

PERFORMANCE ANALYSIS OF PELTON TURBINE

SUBMITITED IN PARTIAL FULFILLMENT FOR THE AWARD OF DEGREE OF

BACHELOR OF TECHNOLOGY

IN

MECHANICAL ENGINEERING

FROM

JIS COLLEGE OF ENGINEERING

MECHANICAL ENGINEERING DEPARTEMENT

PROJECT REPORT

On

Performance Analysis of Pelton Turbine

Guide- Mr. Dipak Shaw

(Prof. Mechanical Engineering Dpt)

By: Anurag Kumar Poddar (12300711007)

Anurag_kumar_Poddar

Forth Year

Mechanical Engineering Departement

JIS OLLEGE OF ENGINEERING

WEST BENGAL OF TECHNOLOGY

Acknowledgement

I would like to take this opportunity to express my sincere thanks to all those without whose support this project could not have been completed.

This project work was a footstep in my career plan, which we did successfully only because of immense support & encouragement from a lots seniors.

I am grateful to MR. DIPAK SHAW for allowing us to take the project in the topic Performance analysis of Pelton Turbine. He has given us the knowledge about all topics that was helpful for our project and we are special thankful to the Lab Assistant for support and help us to do the above so. I am thankful to all for giving his valuable time and help to do our project successful.

I had a wonderful and rich learning experience and I am sure the knowledge I gained here will hold me in good stead in the future.

THANK YOU

ANURAG KUMAR PODDAR

Anurag_Kumar_Poddar

CERTIFICATE

It is hereby certify that Mr. ANURAG KUMAR PODDAR, ASHISH ANAD SINGH, KRISHNA CHANDRA, KUNDAN SAGAR, MANISH KUMAR, RAHUL GUPTA, RAJAT KANTI PANJA, SANDEEP KUMAR THAKUR,B.Tech – IVth Year, Mechanical Engineering, from JIS College of Engineering,Kalyani, West Bengal has completed the project with topic in PERFORMANCE ANALYSIS OF PELTON TURBINE .they have completed project and practical work with full of experience and knowledge.

The matter presented in this Report has made and submitted by them.In this training duration they were very sincere and hard working under my guidance.

Anurag_Kumar_Poddar

Date: 12th May 2015

MR. DIPAK SHAW

(PROF. MECHANICAL ENGEENIRING DEPT.)

JIS College of Engineering

Kalyani

ABSTRACT

Performance Analysis of Pelton Turbine-

To know about the effects on pelton turbine due to change in inlet and outlet pressure of orifice. To know what are the effects of load when it increases slightly and what are the effects over speed of rotation. How to maintain the speed the speed and efficiency.

To know briefly about turbines, its types, its working principle, maintenances, speed regulation and its analysis. To know about Impulse Turbine, its working principle and analysis.

To calculate efficiency of pelton turbine with respect to rotor speed and head and plot graph between load vs efficiency, rotor speed vs efficiency ad load vs rotor speed. Draw the graphs in MATLAB and analise it.

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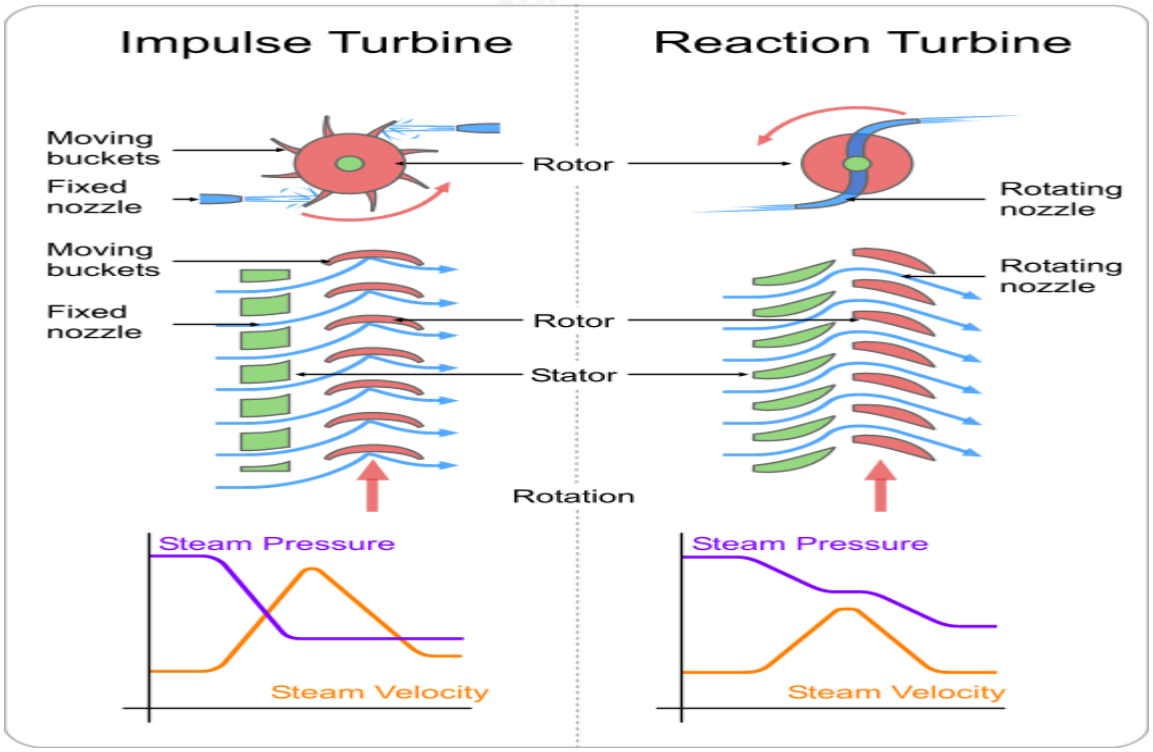
INTRODUCTION

A Turbine is a device that extracts energy from fluid flow and uses it to do mechanical work on a rotating output shaft. Its modern manifestation was invented by Sir Charles Parson in 1884.Basically Steam Turbine has two type, one is Impulse and second is reaction (Depending upon types of blade used and method of energy transfer from fluid to the rotor wheel).

A Impulse Turbine is a type of turbine where all pressure drops of steam occur in the nozzles and there is no pressure drop as steam flows through the passage between two blades.

A Reaction Turbine is a type of turbine where pressure drop occurs both in nozzles or the fixed row of blades, as well as in the moving row of blades, since the blade channels are also of the nozzle shape.



The Pelton Turbine is an impulse type water turbine.The Pelton wheel extracts energy from the impulse of moving water.The inlet pressure of orifice,outlet pressure of orifice, inlet pressure of turbine and rotational speed changes according to the variation on load over it.

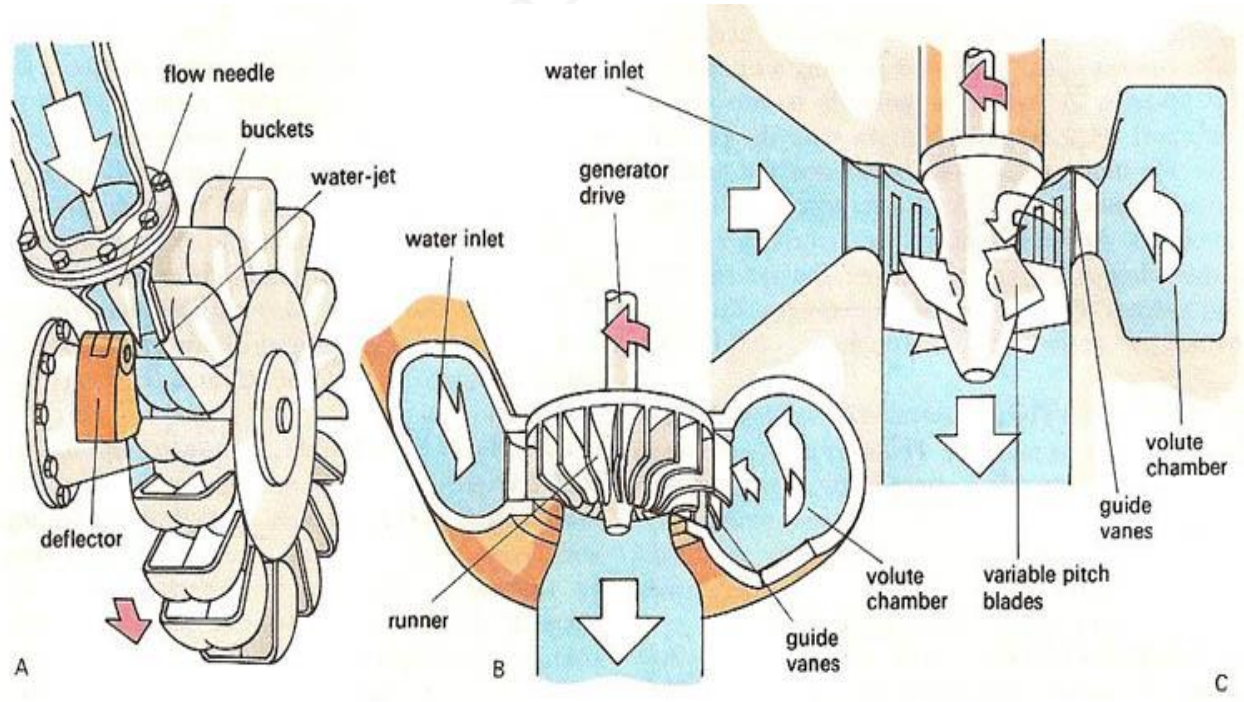
There are many losses taking place during the expansion within a Impulse Turbine, these are internal and external loss. The internal losses are (i)losses in regulating valves (ii)Nozzel friction losses (iii) Blade friction losses (iv) Disc friction losses (v) partial admission losses (vi) gland leakage losses (vii)residual velocity losses (viii)carry over losses . The external losses are the losses in the bearing and governing mechanism which can be reduce by improving lubrication systems.

Certain control and supervisory instruments are provided for the safe and effective operation of a Impulse Turbine like pressure gauge,thermometers, flow meter, watt meter, volt meter, ammeter, handwheel, Governor and a trip lock lever. The control of a turbine with a governor is essential, as turbines need to be run up slowly to prevent damage and some applications (such as the generation of alternating current electricity) require precise speed control.

WORKING PRINCIPLE

Working principle of Pelton turbine is simple. When a high speed water jet injected through a nozzle hits buckets of Pelton wheel; it induces an impulsive force. This force makes the turbine rotate. The rotating shaft runs a generator and produces electricity.

Nozzles direct forceful, high-speed streams of water against a rotary series of spoon-shaped buckets, also known as impulse blades, which are mounted around the circumferential rim of a drive wheel—also called a runner. As the water jet impinges upon the contoured bucket-blades, the direction of water velocity is changed to follow the contours of the bucket. Water impulse energy exerts torque on the bucket-and-wheel system, spinning the wheel; the water stream itself does a "u-turn" and exits at the outer sides of the bucket, decelerated to a low velocity. In the process, the water jet's momentum is transferred to the wheel and thence to a turbine. Thus, impulse energy energy does work on the turbine.



DESIGN FOR PELTON TURBINE

NUMBER OF JETS

Generally a pelton wheel has one nozzle or one jet. However a number of nozzles may be employed when more power is to be produced with the same wheel .The nozzles are spaced evenly around the same runner.

Theoretically 6 nozzles can be used with one pelton wheel.Practical considerations, however, limit the use of not more than 2 jets per runner for a vertical runner and not more than 4 per runner when it is in horizontal position

NUMBER OF BUCKET

The number of buckets should be few as possible so that there is little loss due to friction.

The jet of water must be fully utilized so that no water from the jet goes waste i.e no water escapes without striking the buckets .

$$Z=15+D/2d$$

DEPTH AND WIDTH OF BUCKETS

Width of bucket =5d

Depth of bucket =1.2d

JET RATIO

m represents the ratio of the pitch circle diameter to the jet diameter. (m=D/d)

For maximum hydraulic efficiency ,the jet ratio lies between 11 and 15 and normally a jet ratio of 12 is adopted. A bulky installation results when a larger value of jet ratio is adopted.Its value is nearly equal to 12.

VELOCITY TRIANGLES

The inner velocity triangle is drawn at the splitter and outlet velocity triangle. drawn at the outer edge of the bucket

H=net head acting on the pelton wheel.

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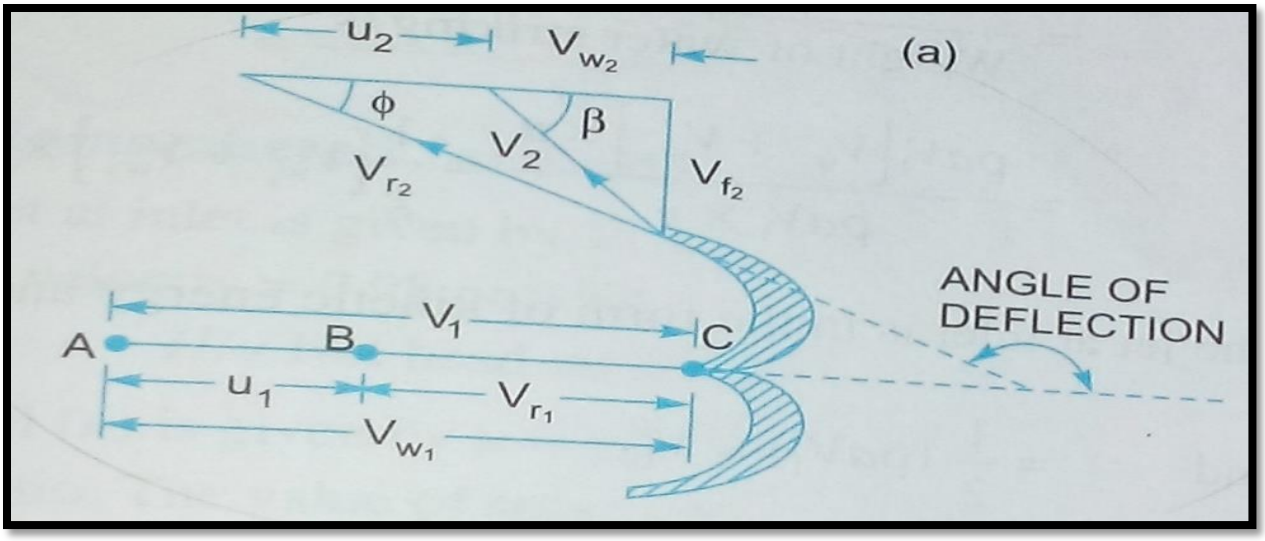

$=H_g - H_f$,

N =speed of the wheel in rpm

D=dia of wheel, d=dia of the jet

THEN V_1=velocity of jet at inlet=sq. root2gH

$u=u_1=u_2=3.14DN/60$

the velocity triangle at inlet will be a straight line where

$V_{r1}=V_1-U_1=V_1-u$

$V_{w1}=V_1$

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POWER

WATER POWER-

The power supplied by the water jet is called as water power, WP.

Power supplied by jet=wQH=pgQH , where Q is the discharge supplied by water jet in cubic meter per sec,and H is the net or effective head in .

BUCKET POWER-

The power developed at the turbine buckets is called as bucket power,BP.

The power delivered by the bucket wheel is

$=m(V_{u1}+V_{u2})u$

EFFICIENCY

OVERALL EFFICIENCY

=power available at the turbine shaft/power available from the water jet

$n_{0=}$ shaft power/water power=**P/wQH**

HYDRAULIC EFFICIENCY

=work done per sec/k.e per second

$\mathbf{n_h=((V_{u1}+V_{u2})u)/gH}$

VOLUMETRIC EFFICIENCY

=Volume of water actually striking the buckets/total water supplied by the jet

$\mathbf{=n_v \ =Q_a/Q=Q-q/Q}$

MECHANICAL EFFICIENCY

=shaft power/water power=$\mathbf{P/wQ_aH}$

Now from hydraulic efficiency

Hydraulic efficiency=work done per second/k.E of jet per second

$=paV1*(V_{w1} + V_{w2})u / 1/2paV_1^2*(V_1)^2$

$=2*(V_{w1} + V_{w2})u/ V_1^2$

By velocity triangles formula, we have hydraulic efficiency=

$=2*[V_1+ (V_1 –u)\cos\omega-u]*u/ V_1^2$

$=2*[(V_1-u)+ (V_1 - u)\cos\omega]*u/ V_1^2$

$=2* (V_1-u)[(1 + \cos\omega]*u/ V_1^2$

=the efficiency will be maximum for a given value of V1 when

D(efficiency)/ d(u) =0

Or, $d\{2* (V_1-u)[(1 + \cos\omega]*u/ V_1^2\} = 0$

$[1 + \cos\omega]*u/ V_1^2*d(2uV1 – 2u^2) = 0$

Or, $d(2uV1 – 2u^2)/ d(u)= 0$

Or, $2V1 -4u =0$. Or, $u= V1/2$

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Efficiency will be obtained by substituting the value of u in efficiency equation

$$\text{Max efficiency} = 2 * (V1 - V1/2)(1 + \cos\varphi) * V1^2 / V1^2$$

$$= 2 * V1/2 * (1 + \cos\varphi) * V1^2 / V1^2$$

$$= (1 + \cos\varphi) / 2$$

We can also check the relation graph between φ and maximum efficiency.

PERFORMANCE ANALYSIS OF PELTON TURBINE

We measured inlet nozzle pressure ,outlet nozzle pressure ,inlet turbine pressure and rotational speed of turbine in pelton turbine in our Mechanical engineering department. We have taken data with the refrences of two type of load, first one is desd load and second is spring load. In dead load we have taken data freom zero load and after it we have taken the spring load condition. Pelton turbine in our department-

DATA

DATA 1

LOAD (Kg)	SPRING LOAD (Kg)	TOTAL LOAD (Kg)	INLET ORIFICE PRESSURE (Kg/cm^2)	OUTLET ORIFICE PRESSURE (Kg/cm^2)	INLET NOZZEL PRESSURE (Kg/cm^2)	SPEED (rpm)	EFFICIENCY ($\eta\%$)
0	0	0	1.8	1.5	1.5	950	44.07
0.25	0	0.25	1.8	1.5	1.5	940	46.71
0.50	0	0.50	1.8	1.5	1.5	925	51.2
0.75	0	0.75	1.8	1.5	1.5	915	52.81
1.00	0	1.00	1.8	1.5	1.5	905	55.21
1.25	0	1.25	1.8	1.5	1.5	890	58.48
1.50	0	1.50	1.8	1.5	1.5	875	61.89
1.75	0	1.75	1.8	1.5	1.5	865	64.30
2.00	0.25	2.25	1.8	1.5	1.5	853	66.23
2.25	0.25	2.50	1.8	1.5	1.5	842	68.48
2.50	0.35	2.85	1.8	1.5	1.5	828	71.1
2.75	0.50	3.25	1.8	1.5	1.5	820	75.14
3.00	0.65	3.65	1.8	1.5	1.5	805	78.33
3.25	0.75	4.00	1.8	1.5	1.5	785	81.37
3.50	0.75	4.25	1.8	1.5	1.5	765	84.46
3.75	1.00	4.75	1.8	1.5	1.5	735	85.33
4.00	1.00	5.00	1.8	1.5	1.5	742	87.98
4.25	1.25	5.50	1.8	1.5	1.5	712	90.4
4.50	1.25	5.75	1.8	1.5	1.5	688	91.52
4.75	1.35	5.10	1.8	1.5	1.5	675	92.34
5.00	1.50	6.50	1.8	1.5	1.5	665	93.09
5.25	1.50	6.75	1.8	1.5	1.5	655	94.39
5.50	1.75	7.25	1.8	1.5	1.5	635	95.72

5.75	1.75	7.50	1.8	1.5	1.5	607	96.15
6.00	2.00	8.00	1.8	1.5	1.5	595	96.68
6.25	2.25	8.50	1.8	1.5	1.5	573	96.9
6.50	2.25	8.75	1.8	1.5	1.5	554	97.3

EFFICIENCIES:

(a) Hydraulic Efficiency(η_h) : $\frac{\text{Power Delivered to runner}}{\text{Power Supplied at Inlet}} = \frac{\text{R.P.}}{\text{W.P}}$

Where R.P= Power delivered to runner=$\frac{[V_{W_1} \pm V_{W_2}]u}{g \times H}$ kw

W.P= Power supplied at inlet of turbine and also called water power

$= \frac{W \times H}{1000}$ kW

Here,

W =Weight of Water striking the vanes of the turbine per second

V_{W_1} = Velocity of whirl at inlet= $\sqrt{2Hg}$

V_{W_2} = Velocity of whirl at outlet=$V_{r_2} \cos\theta - u$

V_{r_1} =Relative velocity of jet and plate at inlet=$V_{W_1} - u$

From the velocity triangle at outlet we have V_{r_1}=V_{r_2}

u= Tangential velocity of vane= $\frac{\pi DN}{60}$

H=Net head on the Turbine

Now,

(1) Load=0

Jet pressure=1.5Kg/cm^2=15 cm

Velocity of whirl at inlet (V_{W_1}) = V = $\sqrt{2Hg}$ =$\sqrt{2 \times 9.81 \times 15}$ =17.15m/s

Tangential velocity of vane (u)= $\frac{\pi DN}{60}$=$\frac{\pi \times 30 \times 950}{60}$ =14.91m/s

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Relative velocity of jet and plate at inlet (V_{r_1}) =$V_{W_1} - u$ = 17.15-14.91
=2.24m/s

Velocity of whirl at outlet (V_{W_2})= $V_{r_2}\cos\theta - u$=2.24cos 20-14.91
= -12.8m/s

So, the Efficiency (η_h) =$\frac{[V_{W_1} \pm V_{W_2}]u}{g\times H}$ = $\frac{[17.15-12.8]\times 14.91}{9.81\times 15}$ = 44.07%

(2) Load=0.25

Velocity of whirl at inlet (V_{W_1}) = V = $\sqrt{2Hg}$ =$\sqrt{2\times 9.81\times 15}$=17.15m/s

Tangential velocity of vane (u)= $\frac{\pi DN}{60}$=$\frac{\pi\times 30\times 940}{60}$ =14.75m/s

Relative velocity of jet and plate at inlet (V_{r_1}) =$V_{W_1} - u$ = 17.15-14.75
=2.4m/s

Velocity of whirl at outlet (V_{W_2})= $V_{r_2}\cos\theta - u$=2.4cos 20-14.75
= -12.8m/s

So, the Efficiency (η_h) =$\frac{[V_{W_1} \pm V_{W_2}]u}{g\times H}$ = $\frac{[17.15-12.49]\times 14.75}{9.81\times 15}$ = 46.71%

(3)Load=0.50

Velocity of whirl at inlet (V_{W_1}) = V = $\sqrt{2Hg}$ =$\sqrt{2\times 9.81\times 15}$ =17.15m/s

Tangential velocity of vane (u)= $\frac{\pi DN}{60}$=$\frac{\pi\times 30\times 925}{60}$ =14.52m/s

Relative velocity of jet and plate at inlet (V_{r_1}) =$V_{W_1} - u$ = 17.15-14.52
=1.18m/s

Velocity of whirl at outlet (V_{W_2})= $V_{r_2}\cos\theta - u$=1.18cos 20-14.52
= -12.04m/s

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So, the Efficiency) $=\frac{[V_{W_1} \pm V_{W_2}]u}{g \times H} = \frac{[17.15-12.04]\times 14.75}{9.81\times 15} = 51.20\%$

(4)Load=0.75, u=14.36, η_h=52.81%

(5) Load=1.00, u=14.20, η_h=55.2%

(6) Load=1.25, u=13.97, η_h=58.48%

(7) Load=1.50, u=13.73, η_h=61.89%

(8) Load=1.75, u=13.55, η_h=64.30%

(9) Load=2.25, u=13.40, η_h=66.23%

(10) Load=2.50, u=13.22, η_h=68.48

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By above data we have seen that when we apply load on the pelton turbine, the rotational speed of the pelton turbine decreases. And after it if we increase the load value ,the rotational speed of pelton turbine is decreases .On above load and pressure condition of inlet and outlet has seen by us while taking experimental data.In both, dead and spring load conditions there are inlet nozzle pressure, outlet nozzle pressure, inlet turbine pressure and rotational speed .

After that we have taken more data for calculation of hydraulic efficiency

$$\eta(\text{Hydraulic}) = \frac{Power\ delivered\ to\ runner}{Power\ supplied\ at\ inlet} = \frac{\text{RP}}{\text{WP}}$$

Where,

R.P = Runner Power

W.P =Water Power

$$\eta = \frac{[V_{w_1} + V_{w_2}]u}{gH}$$

Where, V_{w_1} = Velocity of whirl at inlet,

V_{w_2} = Velocity of whirl at outlet,

U = Tangential velocity of Vane,

H = Net head on the turbine.

TABLE 2

Load (kg)	Spring Load(kg)	Total Load (kg)	Inlet Orific Pressure	Outlet Orific Pressure	Inlet Nozzel Pressure	Speed	Effiency(η)
0	0	0	1.9	1.6	1.4	942	37.45
0.5	0	0.5	1.75	1.45	1.3	913	35.84
1.0	0	1.0	1.7	1.45	1.3	860	50.81
1.5	0	1.5	1.7	1.4	1.29	826	58.47
2.0	0	2.0	1.7	1.4	1.3	782	69.08
2.5	0	2.5	1.69	1.4	1.28	753	73.42
3.0	0	3.0	1.65	1.4	1.25	718	78.30
3.5	0	3.5	1.62	1.4	1.25	678	84.51
4.0	0	4.0	1.62	1.38	1.22	605	91.98
4.5	0	4.5	1.6	1.38	1.21	597	92.35

CALCULATIONS:-

At load Zero kg,

V_{w_1} = V = $\sqrt{2gH}$ = $\sqrt{2 * 9.81 * 14}$ = 16.57

u = $\frac{\pi DN}{60}$ = $\frac{3.14*30*942}{60}$ = 14.78

V_{r_1} = 16.57 – 14.78 = 1.79

V_{w_2} = 1.79 cos 20 – 14.78 = -13.09

η=$\frac{16.57+(-13.09)78}{9.81*14}$ = 37.45

At load 0.5 kg,

V_{w_1} = V = $\sqrt{2gH}$ = $\sqrt{2 * 9.81 * 13}$ = 15.97

U = $\frac{\pi DN}{60}$ = $\frac{3.14*30*913}{60}$ = 14.33

V_{r_1} = 15.97 – 14.33 = 1.64

V_{w_2} = 1.64 cos 20 – 14.33 = -12.78

η=$\frac{15.97+(-12.78)14.33}{9.81*13}$ = 35.84

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At load 1kg,

V_{w_1} = V = $\sqrt{2gH}$ = $\sqrt{2 * 9.81 * 13}$ = 15.97

U = $\frac{\pi DN}{60}$ = $\frac{3.14*30*860}{60}$ = 13.50

V_{r_1} = 15.97 – 13.50 = 2.47

V_{w_2} = 2.47 cos 20 – 13.50 = -11.17

η=$\frac{15.97+(-11.17)13.50}{9.81*13}$ = 50.81

At load 1.5 kg,

V_{w_1} = V = $\sqrt{2gH}$ = $\sqrt{2 * 9.81 * 12.9}$ = 15.90

U = $\frac{\pi DN}{60}$ = $\frac{3.14*30*826}{60}$ = 12.96

V_{r_1} = 15.90 – 12.96 = 2.94

V_{w_2} = 2.94 cos 20 – 12.96 = -10.19

η=$\frac{15.90+(-10.19)12.96}{9.81*12.9}$ = 58.47

At load 2.0 kg,

V_{w_1} = V = $\sqrt{2gH}$ = $\sqrt{2 * 9.81 * 13}$ = 15.97

U = $\frac{\pi DN}{60}$ = $\frac{3.14*30*782}{60}$ = 12.27

V_{r_1} = 15.97 – 12.27 = 3.7

V_{w_2} = 3.7cos 20 – 12.27 = -8.79

η=$\frac{15.97+(-8.79)12.27}{9.81*13}$ = 69.08

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DATA 3

LOAD (Kg)	SPRING LOAD (Kg)	TOTAL LOAD (Kg)	INLET ORIFICE PRESSURE (Kg/cm^2)	OUTLET ORIFICE PRESSURE (Kg/cm^2)	INLET NOZZEL PRESSURE (Kg/cm^2)	SPEED (rpm)	EFFICIENCY (η%)
0	0	0	1.9	1.6	1.4	942	37.45
0	0	0	1.9	1.6	1.4	942	37.45
0	0	0	1.9	1.6	1.4	942	37.45
0	0	0	1.9	1.6	1.4	942	37.45

0	0	0	1.9	1.6	1.4	942	37.45
0.5	0	0.5	1.7	1.4	1.39	916	43.44
1.00	0	1.00	1.68	1.35	1.39	860	57.85
1.50	0	1.50	1.68	1.35	1.38	824	65.20
2.00	0	2.00	1.68	1.35	1.38	780	73.70
2.50	0	2.50	1.68	1.34	1.38	752	78.61
3.00	0	3.00	1.65	1.34	1.38	700	85.97
3.50	0	3.50	1.62	1.34	1.38	643	91.94
4.00	0	4.00	1.62	1.34	1.38	603	94.74
4.50	0	4.50	1.6	1.34	1.38	595	95.14

At load 0 kg,

V_{w_1} = V = $\sqrt{2gH}$ = $\sqrt{2 * 9.81 * 14}$ = 16.57

U = $\frac{\pi DN}{60}$ = $\frac{3.14*30*942}{60}$ = 14.78

V_{r_1} = 16.57 – 14.78 = 1.79

V_{w_2} = 1.79cos 20 – 14.78 = -13.09

η=$\frac{16.57+(-13.09)14.78}{9.81*14}$ = 37.45%

At load 0.5 kg,

V_{w_1} = V = $\sqrt{2gH}$ = $\sqrt{2 * 9.81 * 13.9}$ = 16.51m/sec

U = $\frac{\pi DN}{60}$ = $\frac{3.14*30*916}{60}$ = 14.38

V_{r_1} = 16.57 – 14.38 = 2.13

V_{w_2} = 2.13cos 20 – 14.38 = -12.39

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$\eta = \frac{16.51+(-12.39)14.38}{9.81*13.9}$ = 43.44%

And for last,

At load 4.5 kg,

V_{w_1} = V = $\sqrt{2gH}$ = $\sqrt{2 * 9.81 * 13.8}$ = 16.54m/sec

U = $\frac{\pi DN}{60}$ = $\frac{3.14*30*916}{60}$ = 14.38

V_{r_1} = 16.54 – 14.38 = 2.16

V_{w_2} = 2.16cos 20 – 14.38 = -12.37

$\eta = \frac{16.54+(-12.37)14.38}{9.81*13.8}$ = 95.14%

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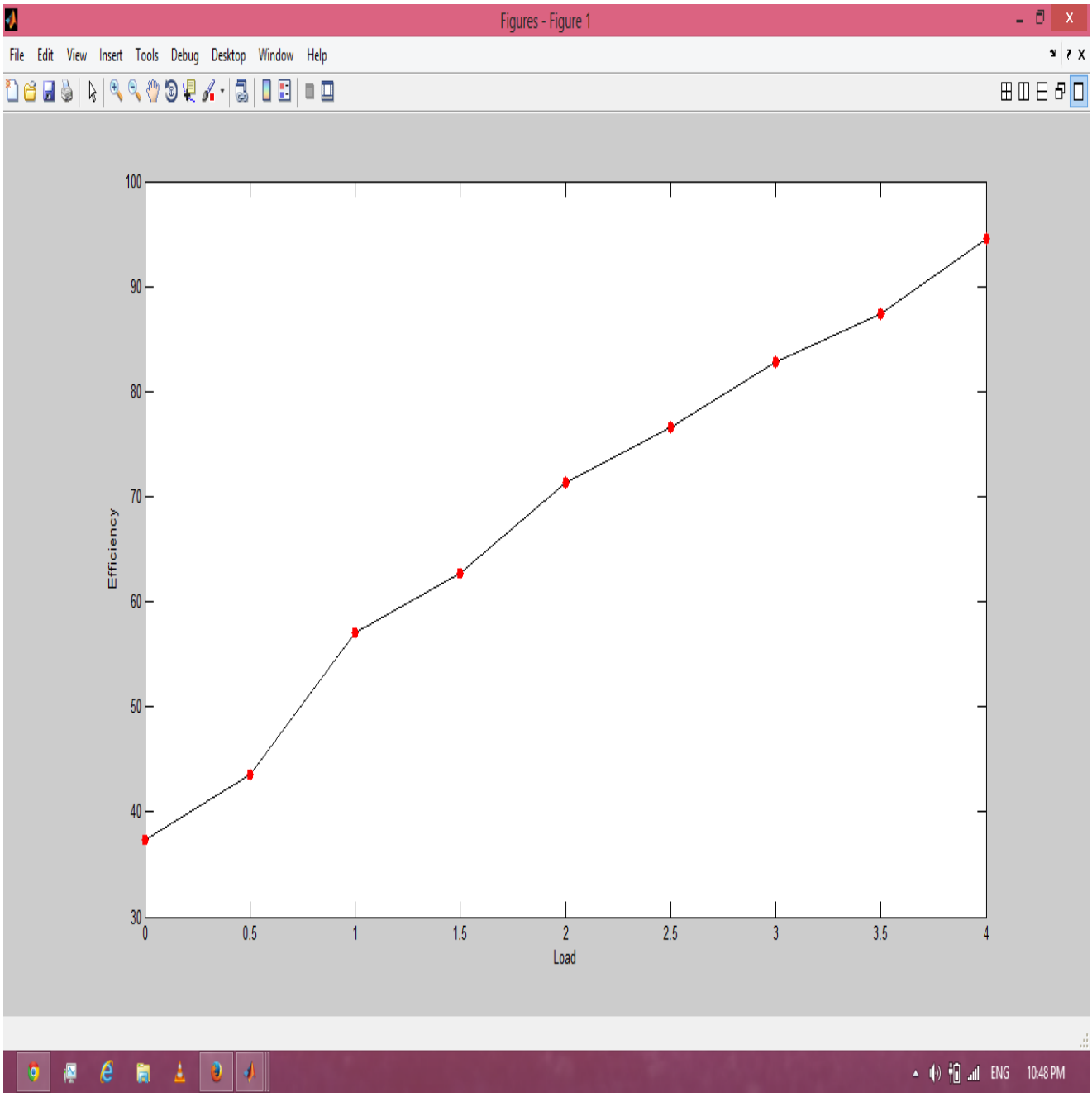
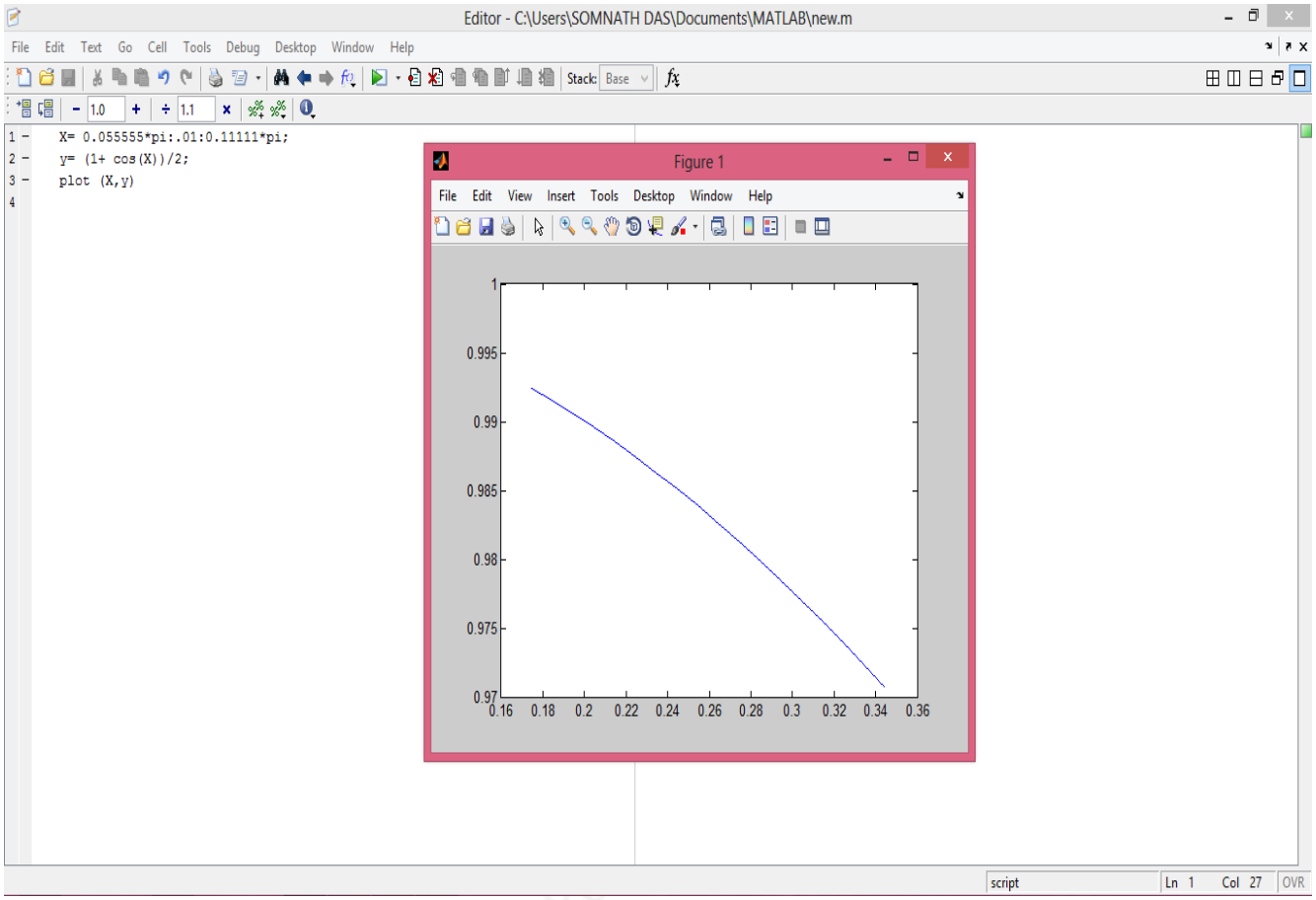
WORKING WITH MATLAB



(EFFICIENCY VS LOAD GRAPH)


Editor - C:\Users\SOMNATH DAS\Documents\MATLAB\new.m
File Edit Text Go Cell Tools Debug Desktop Window Help
Stack: Base
1.0
1.1
X= 0.055555*pi:.01:0.11111*pi;
y= (1+ cos(X))/2;
plot (X,y)
Figure 1
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1
0.995
0.99
0.985
0.98
0.975
0.97
0.16 0.18 0.2 0.22 0.24 0.26 0.28 0.3 0.32 0.34 0.36
script
Ln 1
Col 27
OVR

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CONCLUSION

By performing the above experiment for our project (performance analysis of pelton turbine) .we come to the conclusion that when we increase the load, the rotational speed of turbine accordingly reduced. The data taken during experiment is mentioned on the data sheet.According to inlet nozzle pressure, outlet nozzle pressure, inlet turbine pressure and load condition.

By taking graphs using Matlab we have concluded that efficiency is increasing while decreament in rotor speed in different condition of loads.

maximum efficiency of our turbine is 97%.

Anurag_kumar_Poddar

NOMENCLATURE

- **H_g=** gross head
- H_f= head loss due to friction
- H= net head acting on pelton turbine
- D= dia of wheel
- d= dia of jet
- u= tangential velocity of vane
- V_1= velocity of jet at inlet
- V_{w1}= velocity of whiel at inlet
- V_{w2}= velocity of whirl at outlet
- U_1= tangential velocity of vane at inlet in radial vane
- U_2= tangential velocity of vane at outlet in radial vane
- η_h = hydraulic efficiency of pelton turbine

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Anurag_kumar_Poddar

Anurag_kumar_Poddar